



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Spring 2023

Paper: Solid State Physics - I

Course Code: PHYS-3401

Roll No.

Time: 3 Hrs.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(15x2=30)

- Write the value of coordination numbers for sc, bcc and fcc structure.
- Write down the name of crystal symmetry operations.
- Calculate the packing fraction of fcc lattice.
- Draw (101), (110), (010) and (212) crystal planes.
- Which two crystal structures have similar atomic packing fraction? Write their names.
- How one can differentiate between group velocity and phase velocity? Define each.
- Write dispersion relation for a monoatomic linear lattice and calculate group velocity of the wave at $k = 0$ and at $k = \frac{\pi}{a}$.
- Explain why use of visible light does not show diffraction from crystals?
- What are three mechanisms responsible for repulsive interaction between atoms?
- Define bulk modulus. Derive its equation in terms of volume V and energy U .
- For diatomic lattice, what is the reason of frequency gap at the zone boundary $k_{max} = \pm \frac{\pi}{a}$? Explain.
- Which approximation Debye made in his model to fit theoretical curve (for low temperature limit) with experimental results while explaining lattice heat capacity of solids? Explain.
- Plot the variation of lattice heat capacity with temperature according to classical model.
- Show that the force constant C and potential energy U of two atoms are related with each other as $C = \left(\frac{d^2U}{dR^2} \right)_{R_0}$.
- Explain the terms phonon momentum and crystal momentum.



A

- 1 Coordination number for simple cubic (sc) structure is 6
- 2 Coordination number for body-centered cubic (bcc) structure is 8
- 3 Coordination number for face-centered cubic (fcc) structure is 12

1 The crystal symmetry operations include:

- Identity operation (E)
- Rotation operation (C_n)
- Reflection operation (σ)
- Inversion operation (i)
- Improper rotation operation (S_n)
- Translation operation (t)

(3)-Characteristics of face centered cubic crystal:- In this type of unit cell, in addition to lattice points at the corner, there is one atom at the center of each face. Hence unit cell is non-primitive.

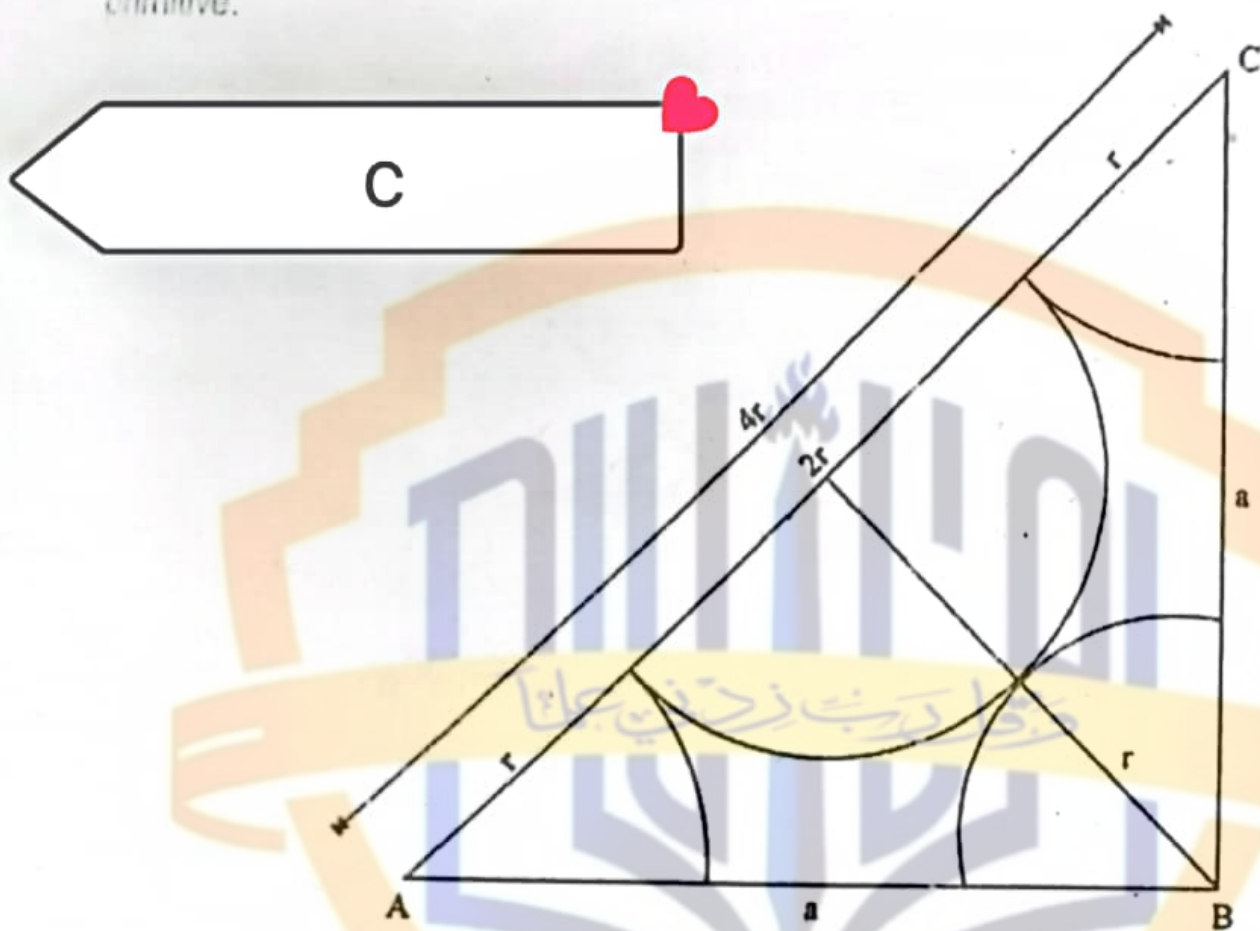


Fig. 1.22

Number of atoms per unit cell is

$$N = \frac{8}{8} + 0 + \frac{6}{2} + 0 = 1 + 3 = 4$$

From fig. 1.22,

$$\begin{aligned} (AC)^2 &= (AB)^2 + (BC)^2 \\ (4r)^2 &= (a)^2 + (a)^2 \Rightarrow (4r)^2 = 2a^2 \\ \Rightarrow a^2 &= \frac{(4r)^2}{2} \Rightarrow a = \frac{4r}{\sqrt{2}} = 2\sqrt{2}r. \end{aligned}$$

Volume of unit cell is $v = a^3 = (2\sqrt{2}r)^3$

Since number of atoms per unit cell are 4, so volume occupied by atoms in unit cell is

Packing fraction is defined as:

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

$$V = 4 \times \left[\frac{4}{3} \pi r^3 \right] = \frac{16}{3} \pi r^3$$

C

$$f = \frac{\text{Volume occupied by all atoms in the cell}}{\text{Volume of unit cell}}$$

$$f = \frac{\frac{16}{3} \pi r^3}{(2\sqrt{2}r)^3} = \frac{\frac{16}{3} \pi r^3}{(2\sqrt{2})^3 r^3} = 0.74$$

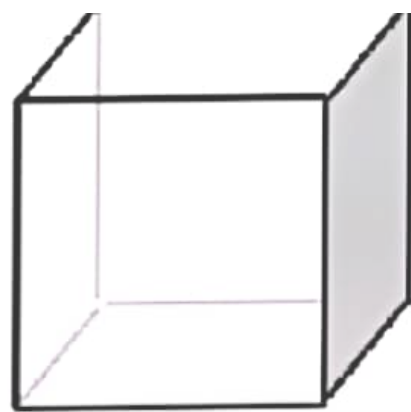
74% filled.
26% empty.

Since number of nearest neighborhood atoms in the cell is 12, so coordination number is "12".

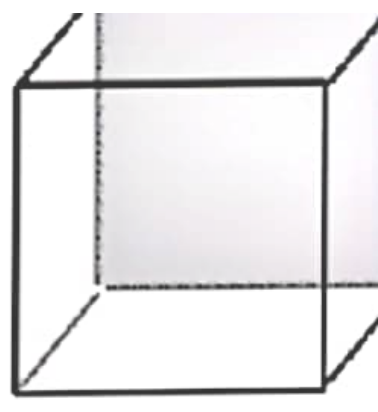
1.8: Index system for crystal planes (Miller indices)



(001)



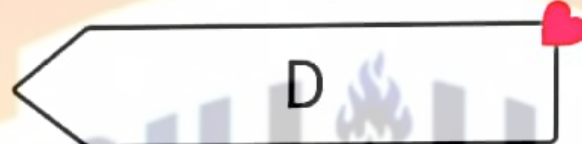
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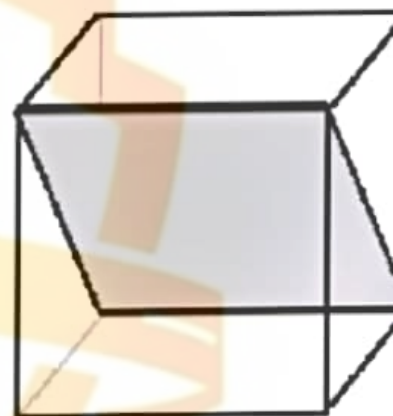
(010)



(101)



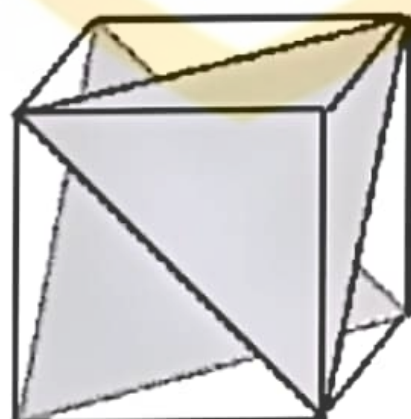
(110)



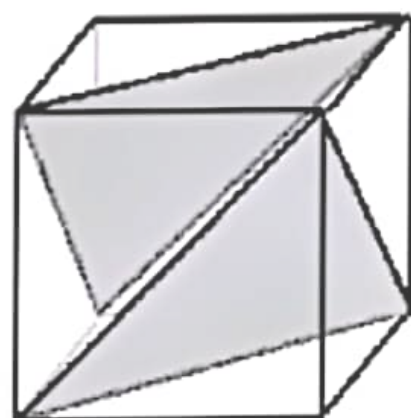
(011)



(111)



(1 $\bar{1}$ 1)



($\bar{1}$ 11)



The face-centered cubic (FCC) and hexagonal close-packed (HCP) crystal structures have similar atomic packing fractions



F

- 1 Group velocity is the velocity at which the envelope of a wave packet propagates. It represents the speed at which the shape of the wave packet moves
- 2 Phase velocity is the velocity at which the phase of a wave propagates. It represents the speed at which the crests of the wave move
- 3 To differentiate between them, consider an example of a wave packet. The group velocity is the speed at which the whole wave packet moves, while the phase velocity is the speed at which individual wave crests move within the packet

This is dispersion relation for monatomic linear lattice.

When $k = 0$, $\omega = 2\sqrt{\frac{C}{m}} \times 0 = 0$, when $k = \frac{\pi}{a}$, $\omega = 2\sqrt{\frac{C}{m}} \times 1 = 2\sqrt{\frac{C}{m}}$ ✓

When $k = \frac{2\pi}{a}$, $\omega = 2\sqrt{\frac{C}{m}} \times 0 = 0$ ✓

The dispersion curve for propagation of wave in monatomic linear lattice is shown in fig. 4.3. It is seen that curve is periodic with a period $\frac{2\pi}{a}$ and ω has maximum value when $k = \frac{\pi}{a}$.

Differentiating dispersion relation w. r. t. to k ,

$$\frac{d\omega}{dk} = 2\sqrt{\frac{C}{m}} \cos \frac{ka}{2} \times \frac{a}{2} = a\sqrt{\frac{C}{m}} \cos \frac{ka}{2}$$

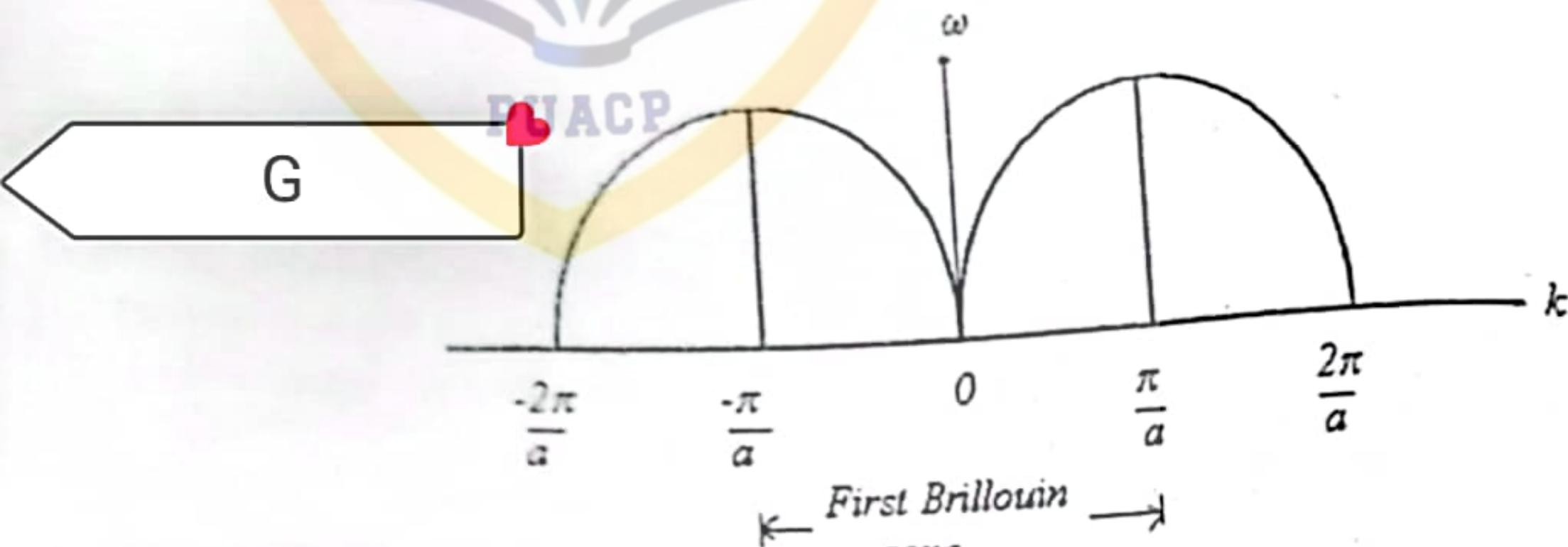
G

But $\frac{d\omega}{dk} = v_g$, group velocity, so

when $k = 0$, $v_g = a\sqrt{\frac{C}{m}}$

$$v_g = a\sqrt{\frac{C}{m}} \cos \frac{ka}{2}$$

At first Brillouin zone, $k = \pm \frac{\pi}{a}$, $v_g = 0$



Q.2.2:- Why visible light cannot be used to study diffraction from crystal?

OR

Why visible light cannot be used to study crystal structure?

Answer:-According to Bragg law,

$$2d \sin \theta = n \lambda$$

The maximum wavelength that can be used for study of crystal structure is $2d$. Since this spacing in crystals is of order of $\text{\AA}$ where as wavelength of visible light is $\sim 5000 \text{\AA}$ which is very large as compared to $2d$. Therefore visible light can not be used for study of crystal structure.

H



- 1 The three mechanisms responsible for repulsive interaction between atoms are:
 - Pauli Exclusion Principle: States that no two electrons in an atom can have the same set of four quantum numbers.
 - Electron-Electron Repulsion: Electrons repel each other due to their negative charges, leading to repulsion between atoms.
 - Electron-Nucleus Repulsion: Electrons are attracted to the positively charged nucleus but also repelled by it due to the like charges, contributing to the repulsive interaction between atoms.

- 1 Phonon momentum refers to the momentum carried by a phonon, which is a quantized lattice vibration in a crystalline solid. Phonons are considered as quasi-particles that represent the collective motion of atoms in a crystal lattice. The momentum of a phonon is related to its wavelength and frequency
- 2 Crystal momentum, on the other hand, is a concept in solid-state physics that describes the momentum of an electron in a crystal lattice. It takes into account the periodicity of the crystal lattice structure. Crystal momentum is a useful concept in understanding electronic band structures and electron behavior in solids